INCH-POUND

MS90707C
16 OCTOBER 2012
SUPERSEDING
MS90707B
25 August 1989

DETAIL SPECIFICATION SHEET

RING, RETAINING, EXTERNAL GRIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-R-21248.

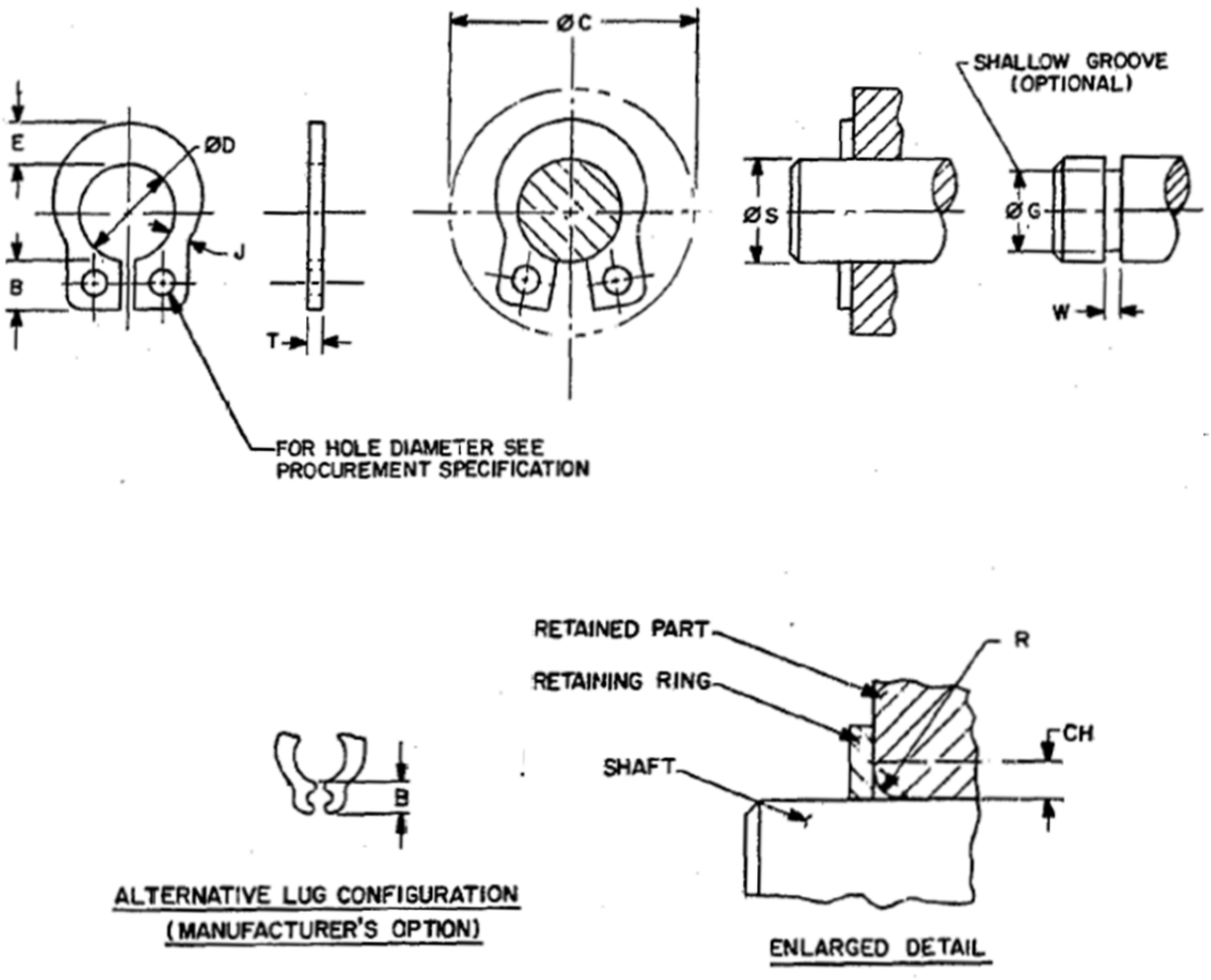


FIGURE 1. RING, RETAINING.

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TABLE I DIMENSIONS.

ØS SHAFT (REF)			ØD FREE		B LUG HEIGHT		E LARGE HEIGHT		J SMALL SECTION HEIGHT		T 1/ THICKNESS		ØC 2/ CLEAR	R 3/ OF RETAINED PART (REF)	CH 3/	ØG SHALLOW GROOVE (OPTIONAL)		W WIDTH	
INCH	MM	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL		MAX	MAX	BASIC	TOL	BASIC	TOL
.094	2.4		.089	+.002	.078	±.008	.045		.028		.025		.26	.042	.025	SIZES .094 THROUGH .187 ARE NOT RECOMMENDED FOR USE WITH GROOVES			
.125	3.2		.120	-.003	.078		.070	±.004	.048	±.004	.025	±.002	.33	.054	.032				
.156	4.0	±.002	.150	+.002	.078	±.003	.079		.051		.025		.36	.059	.035				
.187	4.8		.181	-.004	.097		.086		.052		.035		.44	.063	.038				
.250	6.3		.238		.097		.101		.057		.035		.49	.072	.043	.240	+.0005 -.0015	.041	+.003 -.000
.313	7.9		.298	+.003	.141		.114	±.005	.073	±.005	.042	±.003	.68	.080	.048	.303		.048	
.376	9.5		.354	-.005	.141		.125		.075		.042		.74	.086	.051	.361		.048	
.437	11.1	±.003	.412		.151	±.004	.138		.083		.050		.81	.093	.056	.419	+.001	.056	
.500	12.7		.470		.158		.140		.082		.050		.90	.100	.060	.478	-.002	.056	+.004
.625	15.9		.593	+.004 -.006	.180		.175	±.006	.100	±.006	.062	±.004	1.06	.120	.072	.599		.069	-.000
.750	19.0	±.005	.706		.233		.176		.104		.062		1.32	.125	.075	.718	+.002 -.003	.069	

1/ T - THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e. ±.002 SHOULD BE +.004/-.002.

2/ C - ACTUAL CLEARANCE DIAMETER WHEN THE RING IS SPRUNG OVER THE SHAFT. (DESIGN REFERENCE DIMENSION).

3/ R AND CH - RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. THRUST LOADS FOR RINGS, RETAINING PARTS WITH CORNER RADII OR CHAMFERS, ARE TABULATED ON PAGE 6.

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REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE I, CLASS 7 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:

(a) CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A 568/A568M OR ASTM A684/A684M.

(b) CORRISION RESISTANT STEEL, IN ACCORDANCE WITH SAE AMS 5520 (UNS S15700).

(c) BERYLLIUM COPPER, ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194.

3. HARDNESS

TABLE II. HARDNESS.

ØSHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.094 TO .156	83.5-86 HR15N	82.5-86 HR15N	77-82 HR15N
.187 TO .500	65-69.5 HR30N	63-69.5 HR30N	54-62 HR30N
.625 TO .750	46-51 HRC	44-51 HRC	34-43 HRC

4. PROTECTIVE FINISH OR TREATMENT:

(a). CARBON STEEL - SHALL BE (SEE TABLE III OR IV)

(1) CADMIUM PLATE IN ACCORDANCE WITH SAE AMS-QQ-P-416, TYPE II, CLASS 3 OR ASTM B 696, TYPE II, CLASS 5.
(2) ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn 5 OR ASTM B 695, TYPE II, CLASS 5.
(3) PHOSPHATE COAT IN ACCORDANCE WITH MIL-DTL-16232, TYPE Z, CLASS 2.

(b) CORROSION RESISTANT STEEL - SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH SAE AMS2700.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III OR IV.

EXAMPLE: MS90707-1025 IS THE PART NUMBER FOR A CARBON STEEL, CADMIUM PLATE, EXTERNAL GRIP, RETAINING RING FOR USE ON A .250 DIAMETER SHAFT.

TABLE III. DASH NUMBERS FOR MS90707

Ø S SHAFT (REF)	1/ CARBON STEEL CADMIUM PLATE	1/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT	STEEL, CORROSION RESISTANT	1/ BERYLLIUM COPPER
	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
.094	-1009	-2009	-3009	-4009	-5009
.125	-1012	-2012	-3012	-4012	-5012
.156	-1015	-2015	-3015	-4015	-5015
.187	-1018	-2018	-3018	-4018	-5018
.250	-1025	-2025	-3025	-4025	-5025
.313	-1031	-2031	-3031	-4031	-5031
.376	-1037	-2037	-3037	-4037	-5037
.437	-1043	-2043	-3043	-4043	-5043
.500	-1050	-2050	-3050	-4050	-5050
.625	-1062	-2062	-3062	-4062	-5062
.750	-1075	-2075	-3075	-4075	-5075

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

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TABLE IV. SUBSTITUTION TABLE, CROSS REFERENCE OF PART NUMBERS

Ø S SHAFT (REF)	INACTIVE CARBON STEEL	SUBSTITUTE 1/ CARBON STEEL CADMIUM PLATE	SUBSTITUTE 1/ CARBON STEEL ZINC COAT	SUBSTITUTE 1/ CARBON STEEL PHOSPHATE COAT
	MS90707	MS90707	MS90707	MS90707
.094	-9	-1009	-2009	-3009
.125	-12	-1012	-2012	-3012
.156	-15	-1015	-2015	-3015
.187	-18	-1018	-2018	-3018
.250	-25	-1025	-2025	-3025
.313	-31	-1031	-2031	-3031
.376	-37	-1037	-2037	-3037
.437	-43	-1043	-2043	-3043
.500	-50	-1050	-2050	-3050
.625	-62	-1062	-2062	-3062
.750	-75	-1075	-2075	-3075

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. RECOMMENDED DESIGN LIMITATIONS AND USAGE.

(a) INTENDED USE- TO PROVIDE SHOULDERS FOR POSITIONING AND RETAINING MACHINE COMPONENTS ON UNGROOVED SHAFTS, TUBES, BOSSES, STUDS, ETC. FRICTION FORCE CAUSED BY HEAVY SPRING PRESSURE OF RING ON SHAFT MAKES FASTENER SECURE AGAINST AXIAL DISPLACEMENT FROM EITHER DIRECTION UNDER MODERATE THRUST OR VIBRATION. THE RINGS ARE ADJUSTABLE ON THE SHAFT AND ARE RE-USABLE FOLLOWING DISASSEMBLY (SEE WARNING NOTE BELOW). THE RINGS WILL WITHSTAND HIGH ROTATIONAL SPEEDS.

LIMITATION ON USE- THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING - RINGS SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE. RINGS SHOULD ONLY BE EXPANDED TO GLIDE SMOOTHLY OVER THE SHAFT.

FOR APPROXIMATE SAFETY RPM LIMITS SEE TABLE V.

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TABLE V. APPROXIMATE SAFETY RPM LIMITS.

Ø SHAFT(INCHES)		.125	.250	.312	.375	.437	.500	.625	.750
CARBON STEEL AND CORROSION RESISTANT STEEL	RPM LIMIT	80,000	77,000	58,000	51,000	44,000	40,000	32,000	25,000
BERYLLIUM COPPER	RPM LIMIT	51,000	49,000	37,000	32,000	28,000	25,000	20,000	16,000

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS ARE LISTED IN TABLE VII.

(c) THE RINGS ARE BASICALLY DESIGNED TO BE USED WITHOUT GROOVES. THE THRUST LOAD CAPACITY OF A GRIP RING ASSEMBLY MAY BE INCREASED SUBSTANTIALLY BY INSTALLING THE RING IN A SHALLOW GROOVE (ONLY FOR SIZES .250 AND OVER). THE STRENGTH OF THE GROOVE WALL THEN BECOMES THE LIMITING FACTOR IN THE ASSEMBLY.

ALLOWABLE LOAD CAPACITY OF GROOVE WALL: (RETAINED PARTS HAVING SHARP CORNERS, OR RADII, OR CHAMFERS UP TO SPECIFIED LIMITS.)

$$P = \frac{C_F \pi S d Y}{F}$$

WHERE:

P = ALLOWABLE COMPRESSION LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY, F= 2 IS RECOMMENDED TO INSURE A SAFE WORKING LOAD.
C_F = CONVERSION FACTOR, C_F = 1 IS RECOMMENDED SINCE CONTACT AREA IN THE GROOVE WALL IS AROUND ENTIRE CIRCUMFERENCE.

(d) WHEN RINGS ARE USED IN GROOVES, THE MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL Z SHALL BE =

TABLE VI. OUTER GROOVE WALLS.

NOMINAL RING SIZE	Z (MIN)
.250 TO .437	.030
.500	.040
.625	.045
.750	.050

(e) DIFFERENTIAL ROTATION =

DIFFERENTIAL ROTATION OF RING AND ADJACENT PART CREATES NO ELEMENT OF RISK IN THE APPLICATION OF THE RING.

(f) IMPACT CAPACITY OF RING ON SHAFTS WITHOUT GROOVES IS MODERATE. IF RINGS ARE USED IN GROOVES, THE IMPACT CAPACITY INCREASES AND THE GROOVE WALL IS THE LIMITING FACTOR IN THE ASSEMBLY.

IMPACT CAPACITY OF GROOVE WALL=

$$I_G = \frac{Pd}{2}$$

WHERE:

P = ALLOWABLE THRUST LOAD (POUNDS)
d = GROOVE DEPTH (INCHES)
I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS)

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(g) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED IS THE SAME AS THE LOAD CAPACITY OF RINGS ABUTTING COMPONENTS WITH SHARP CORNERS AND IS SPECIFED IN TABLE VII BELOW.

TABLE VII. LIMIT LOADS. 1/

NOMINAL RING SIZE	ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITHOUT GROOVE, RETAINED PART SHARP CORNERED RADIUSED OR CHAMFERED	
	CARBON STEEL OR CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.094	8 LB	6 LB
.125	10 LB	8 LB
.156	12 LB	9 LB
.187	20 LB	15 LB
.250	23 LB	17 LB
.313	25 LB	19 LB
.376	30 LB	23 LB
.437	40 LB	30 LB
.500	45 LB	32 LB
.625	60 LB	45 LB
.750	65 LB	47 LB

1/ LOAD VALUES ARE BASED UPON THE RING INSTALLED ON A SOFT LOW CARBON STEEL SHAFT. FOR HARDENED SHAFTS (50 HRC MIN) THE ALLOWABLE LOAD SHOULD BE REDUCED BY 50 PERCENT.

4. CHANGES FROM PREVIOUS ISSUE. MARGINAL NOTATIONS ARE NOT USED IN THIS REVISION TO IDENTIFY CHANGES WITH RESPECT TO THE PREVIOUS ISSUE DUE TO THE EXTENT OF THE CHANGES.

MILITARY INTEREST

Custodians:
Army - AR
Navy - OS
Air Force - 99

Preparing activity:
DLA - IS

(Project 5325-2012-017)

Review activities:
Army – AT, AV, GL, MI
Air Force - 71

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